



Pharm*Achieve*

DRUGS IN PREGNANCY

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ACEi	Angiotensin-Converting Enzyme Inhibitors
ARBs	Angiotensin II Receptor Blockers
BID	Bis In Die (Twice Daily)
CNS	Central Nervous System
CYP	Cytochrome P450
DVT	Deep Vein Thrombosis
G6PD	Glucose-6-Phosphate Dehydrogenase Deficiency
GERD	Gastroesophageal Reflux Disease
GFR	Glomerular Filtration Rate
GI	Gastrointestinal
IM	Intramuscular
IV	Intravenous
MS	Multiple Sclerosis
PO	Per Os (Oral)
SMX-TMP	Sulfamethoxazole/Trimethoprim

MAIN OBJECTIVES

- 1 Understand principles of medication management during pregnancy and breastfeeding
- 2 Provide drug recommendations for patients who are pregnant and breastfeeding

INTRODUCTION

Teratogen: any agent that can cause malformation or impairs the proper development of an embryo or fetus

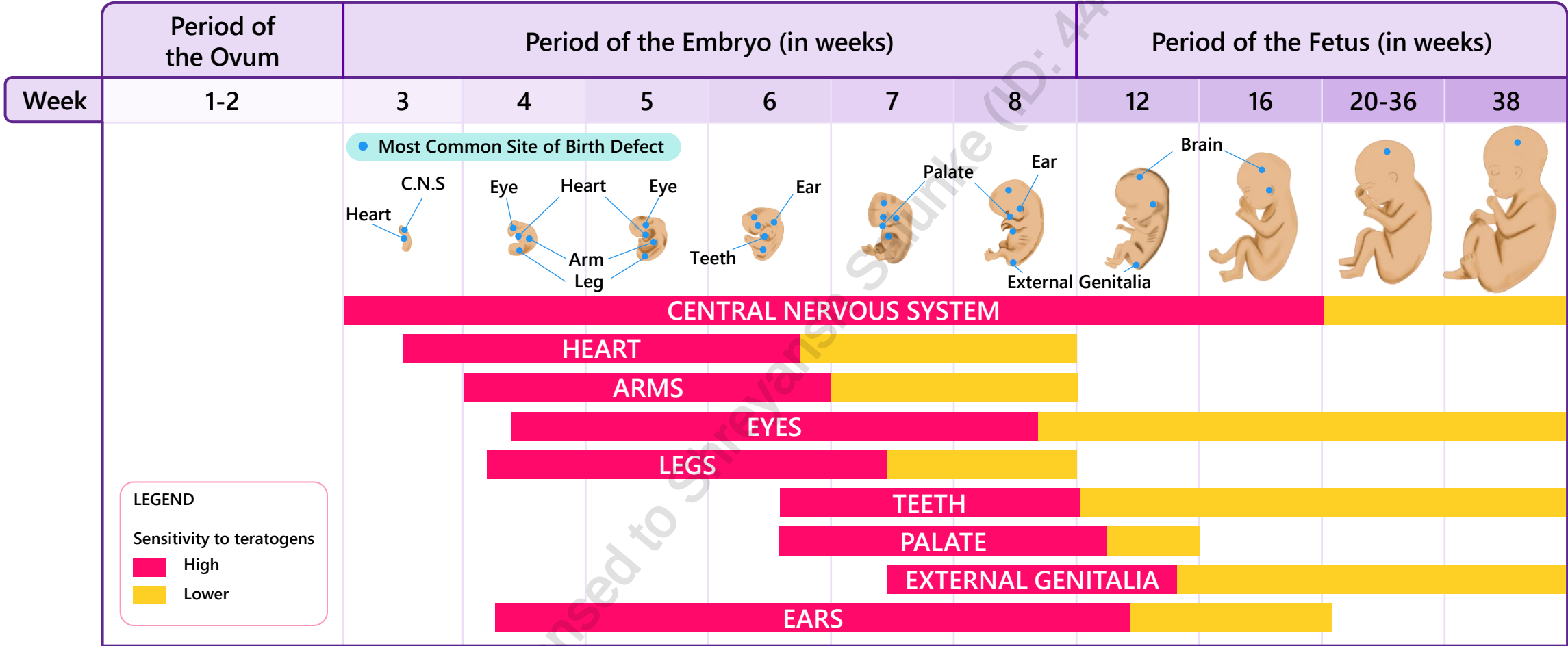
Pharmacokinetics in Pregnancy

- ↑ **water content** in the body
 - Need for larger doses of water-soluble drugs (e.g. aminoglycosides)
- ↑ **cardiovascular output**
- ↑ **GFR** and ↑ **renal blood flow**
- ↓ **serum protein binding**
- ↓ **intestinal motility** (due to progesterone-mediated smooth muscle relaxation)
- ↓ **gastric acidity** thus ↑ **gastric pH**

Placental Drug Transfer

- The **placenta** acts as an **intravenous portal** for entry of drugs into the fetus
- Several **drug properties** increase the chance of placental transfer
 - **High fat-solubility**
 - **Low degree of ionization**
 - **Low degree of protein binding**
 - **Low molecular mass** (<500 Da rapid transit, 500-1000 Da slower transit)

CRITICAL PERIODS IN PRE-NATAL DEVELOPMENT



The greatest sensitivity to teratogens is during organogenesis which lasts about 6 weeks while teratogens during the fetal period are more likely to cause problems with organ function or to stunt growth rather than causing structural damage

PROVEN TERATOGENS



ACEi/ARBs

- Intrauterine renal insufficiency, neonatal hypotension, oliguria with renal failure, hyperkalemia, intrauterine growth restriction, oligohydramnios, prematurity, fetal death

Antineoplastic Agents

- Exposure in 1st trimester leads to malformations and miscarriages

Carbamazepine

Exposure in 1st trimester: neural tube defects (risk: 1% which is 10 times greater than the baseline risk), cardiovascular malformations

Cocaine

- Prematurity, death, microcephaly, limb defects, and urinary tract malformations. Note: cocaine use and abuse is linked to alcohol, drug abuse, & smoking, but epidemiological studies show that cocaine is not structurally a teratogen

Corticosteroids (systemic)

Exposure in 1st trimester: oral clefts (risk: <1%)

Mycophenolate Mofetil

Exposure in 1st trimester: CNS, facial (ear, eye, oral cleft), cardiac, GI, skeletal, and urogenital malformations

PROVEN TERATOGENS



Warfarin

- Exposure in 1st trimester: fetal warfarin syndrome, restriction in intrauterine growth, delay in CNS development, hearing and vision impairment, high rates of miscarriage, hemorrhage

Ethanol

- Fetal alcohol spectrum disorders, neuro-developmental disorders, cleft palate, cardiac malformations

Folic Acid Antagonists

- Abnormalities in CNS development and cranial ossification, intrauterine growth restriction linked to exposure after first trimester. Maternal dose of >10 mg/week will induce defects with methotrexate

Phenytoin

- Cardiovascular, craniofacial, and CNS malformations

Lithium

- Small risk of cardiac malformations for doses >900 mg/day. (e.g. Ebstein anomaly – structural cardiac defect)

Misoprostol

- Exposure in 1st trimester: CNS abnormalities, Moebius sequence (facial palsy). Absolute risk 1-2%. Note: misoprostol induces uterine contractions which results in fetal malformations

PROVEN TERATOGENS



Retinoids and high doses of Vitamin A

- Craniofacial, cardiac, and CNS malformations, miscarriage (risk: 28% anomalies, 40% for miscarriage)

Tetracyclines

- Discolouration of teeth (after 17 weeks gestation), crowns of permanent teeth (close to term) may become stained. Note: doxycycline has a lower risk of enamel staining

Thalidomide

- Limbs, ears, cardiac, and GI malformations, embryopathy. Note: single dose of less than 1 mg/kg is teratogenic. Teratogen potential effects as early as: day 34–50 after the beginning of the last menstrual period

Topiramate

- Exposure during 1st trimester: oral clefts (risk: <1-2%) Note: monotherapy with topiramate <100 mg poses no teratogenic risk

Valproic Acid

- Exposure during 1st trimester: neural tube defects, cardiac, craniofacial, CNS malformations, hypospadias

POSSIBLE TERATOGENS



Diazepam

- Smaller studies noticed a small risk between cleft lip/palate and maternal diazepam, intake however larger studies could not confirm this association

Fingolimod (used in MS)

- Small studies show possible risk of increased congenital malformations (musculoskeletal abnormalities, renal, and congenital heart disease)

Fluconazole (high dose)

- Case reports of cardiac, skeletal, and facial abnormalities

POSSIBLE TERATOGENS



Methimazole

- Case reports of scalp defects; epidemiological reports indicating embryopathy

Penicillamine (high dose)

- Cutis laxa - connective tissue anomalies

Sulfamethoxazole/Trimethoprim (SMX-TMP)

- Possible risk of oral clefts, neural tube defects, and cardiac anomalies (1st term exposure) however supplementing with folic acid may reduce risks; administering close to term is associated with a risk of neonatal hyperbilirubinemia and kernicterus

Statins

- Retrospective reports indicates an increase in limb and CNS anomalies however this could not be confirmed by cohort studies and meta-analysis



Pharm*Achieve*

DRUGS IN BREASTFEEDING

Qualifying Exam Preparatory Course

FACTORS TO CONSIDER

Drug characteristics

Route of administration of drug to mother

- Oral medications: Undergoes first-pass metabolism before entering the blood stream
- IV medications: Bypasses the barriers of the GI tract to enter the bloodstream and the milk
- IM medications: Bypasses barrier of the GI tract and enters the bloodstream through blood vessels surrounding the muscles
- Topical medications: Low systemic absorption

CYP 2D6 substrates

- Codeine → can lead to increased active metabolite levels in infants causing toxicity

Baby's age, maturity, weight, and health status

- GI transit time and flora, rate of metabolism and excretion

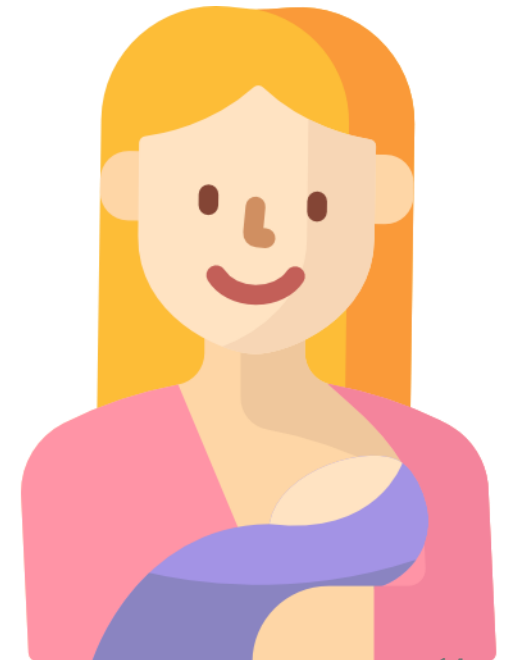
Frequency and volume of milk feedings

Premature/low birth weight infants or G6PD deficiency

- Infants with G6PD deficiency may develop hemolytic anemia if exposed through breastmilk

CHARACTERISTICS OF DRUGS ENTERING BREASTMILK

- Low plasma protein binding
- Non-ionized
- Low molecular weight
- Lipophilic
- Weakly alkaline



CLINICAL PEARLS

- Choose a drug with the shortest half-life, highly protein-bound, poor oral absorption, lowest lipophilicity & with the most data for use in infants & breastfeeding
- Use the lowest effective dose for the shortest period of time
- Take the drug immediately after feeding the infant to minimize drug exposure to infant
- May need to suggest expressing (e.g. pumping) milk if there is a concern with the drug in terms of side effects/harm to the infant
- Encourage breastfeeding and discourage only when there is substantial evidence that the drug is harmful to the infant & there are no other safer therapeutic equivalents
- Take drugs if absolutely necessary during pregnancy & breastfeeding based on risk-benefit assessment. If therapy can be delayed, it is preferred



Only a few drugs are absolutely contraindicated in breastfeeding



DRUGS INCOMPATIBLE WITH BREASTFEEDING

Alcohol and Tobacco

Anticonvulsants (phenobarbital, ethosuximide, primidone)

Antineoplastics

Combined Oral Contraceptives (only use after 6 weeks post-partum)

Ergot Alkaloids

Radiopharmaceuticals e.g. iodine-131 (can lead to hypothyroidism in the infant)

Lithium Carbonate

Street Drugs and Cannabis

Sympathomimetics (pseudoephedrine)

Tetracyclines (staining of immature teeth & change in bone growth with chronic use)

ADDITIONAL RESOURCES



LactMed®

The National Institute of Health National Library of Medicine searchable database on information on drugs and breastfeeding provides the most up-to-date online information on drugs and breastfeeding; type in generic name of medication; analysis and references provided



The Centre IMAGe

Teratogen information service dedicated to offer consultations to healthcare professionals on the use of medications during pregnancy and breastfeeding. Developed in partnership with the Faculty of Pharmacy of the Université de Montréal and the Pharmacy Department of CHU Ste-Justine



MotherToBaby®

Service of the non-profit Organization of Teratology Information Specialists (OTIS), is the nation's leading authority and most trusted source of evidence-based information on the safety of medications and other exposures during pregnancy and while breastfeeding



Drugs in Pregnancy and Lactation (BRIGGS)

Provides information on studies in pregnancy and lactation as well as general recommendations on drug use



Academy of Breastfeeding Medicine® (ABM)

Worldwide organization of physicians dedicated to the promotion, protection and support of breastfeeding and human lactation. Their webpage has a selection of protocols of interest to pharmacists on galactagogues, analgesia, antidepressants, contraception and breastfeeding and the drug dependent woman



InfantRisk Center

The world's leading research center for medication safety during pregnancy and lactation. It is a world-wide call center located in the Texas Tech University Health Sciences Centre, School of Medicine, Department of Pediatrics, in Amarillo



Compendium of Pharmaceuticals and Specialties (CPS)

Provides an overview of drug use in pregnancy and breastfeeding. It is not intended to be a comprehensive review rather a quick summary

TAKEAWAY

A teratogen is any agent that can cause malformation or impairment to the proper development of an embryo or fetus

Drugs should be taken only if necessary during pregnancy and breastfeeding, and the decision should be based on a risk-to-benefit assessment (delay therapy if possible)

The greatest sensitivity to teratogens is during organogenesis which lasts about 6 weeks

Teratogens during the fetal period are more likely to cause problems with organ function or to stunt growth

In breastfeeding mothers, drugs with a short half-life, highly protein-bound, poor oral absorption, low lipophilicity, and evidence for use in infants and breastfeeding are preferred

Encourage breastfeeding (unless a drug is harmful) and take the drug immediately after feeding the infant

CASE

SU is a 32-year-old female who has just found out she is pregnant and is currently at 5 weeks' gestation. She presents to the pharmacy for a medication review. Her past medical history is significant for generalized tonic-clonic seizures, hypertension, migraine headaches, gastroesophageal reflux disease (GERD), and deep vein thrombosis (DVT). Her current medications include carbamazepine 100 mg PO BID, lisinopril 10 mg PO daily, topiramate 200 mg PO daily, omeprazole 20 mg PO daily, folic acid 5 mg PO daily, enoxaparin 40 mg SC daily, and a prenatal multivitamin supplement PO BID. SU requests further information regarding her medication use in pregnancy as she does not want to risk causing any harm to her baby. She is particularly worried as this is her first pregnancy. She also asks whether her current medications will be safe to continue when she begins to breastfeed.

1. Which of the following agents does NOT need to be switched to a therapeutic alternative in SU?
 - a) Topiramate
 - b) Lisinopril
 - c) Enoxaparin
 - d) Carbamazepine



CASE

SU is a 32-year-old female who has just found out she is pregnant and is currently at 5 weeks' gestation. She presents to the pharmacy for a medication review. Her past medical history is significant for generalized tonic-clonic seizures, hypertension, migraine headaches, gastroesophageal reflux disease (GERD), and deep vein thrombosis (DVT). Her current medications include carbamazepine 100 mg PO BID, lisinopril 10 mg PO daily, topiramate 200 mg PO daily, omeprazole 20 mg PO daily, folic acid 5 mg PO daily, enoxaparin 40 mg SC daily, and a prenatal multivitamin supplement PO BID. SU requests further information regarding her medication use in pregnancy as she does not want to risk causing any harm to her baby. She is particularly worried as this is her first pregnancy. She also asks whether her current medications will be safe to continue when she begins to breastfeed.

2. Which of the following recommendations can be made for SU's generalized tonic-clonic seizures?
- a) Discontinue carbamazepine immediately and titrate levetiracetam to a therapeutic dose
 - b) Titrate lamotrigine and if tolerated, gradually taper down carbamazepine
 - c) Titrate valproic acid and if tolerated, gradually taper down carbamazepine
 - d) Discontinue carbamazepine immediately and continue topiramate



CASE

SU is a 32-year-old female who has just found out she is pregnant and is currently at 5 weeks' gestation. She presents to the pharmacy for a medication review. Her past medical history is significant for generalized tonic-clonic seizures, hypertension, migraine headaches, gastroesophageal reflux disease (GERD), and deep vein thrombosis (DVT). Her current medications include carbamazepine 100 mg PO BID, lisinopril 10 mg PO daily, topiramate 200 mg PO daily, omeprazole 20 mg PO daily, folic acid 5 mg PO daily, enoxaparin 40 mg SC daily, and a prenatal multivitamin supplement PO BID. SU requests further information regarding her medication use in pregnancy as she does not want to risk causing any harm to her baby. She is particularly worried as this is her first pregnancy. She also asks whether her current medications will be safe to continue when she begins to breastfeed.

3. SU develops asymptomatic bacteriuria at 38 weeks' gestation. Which of the following agents is most appropriate to recommend?
- a) Nitrofurantoin 100 mg PO BID for 5 days
 - b) Sulfamethoxazole 800 mg/trimethoprim 160 mg PO BID for 5 days
 - c) Fosfomycin 3 g PO once
 - d) Ciprofloxacin 500 mg PO daily for 3 days



CASE

SU is a 32-year-old female who has just found out she is pregnant and is currently at 5 weeks' gestation. She presents to the pharmacy for a medication review. Her past medical history is significant for generalized tonic-clonic seizures, hypertension, migraine headaches, gastroesophageal reflux disease (GERD), and deep vein thrombosis (DVT). Her current medications include carbamazepine 100 mg PO BID, lisinopril 10 mg PO daily, topiramate 200 mg PO daily, omeprazole 20 mg PO daily, folic acid 5 mg PO daily, enoxaparin 40 mg SC daily, and a prenatal multivitamin supplement PO BID. SU requests further information regarding her medication use in pregnancy as she does not want to risk causing any harm to her baby. She is particularly worried as this is her first pregnancy. She also asks whether her current medications will be safe to continue when she begins to breastfeed.

4. Which of the following sets of drug properties is preferred for women who are breastfeeding?
- a) Low protein-binding, low lipophilicity
 - b) Short half-life, high lipophilicity
 - c) Poor oral absorption, long half-life
 - d) High protein-binding, low lipophilicity



CHANGES LOG

April 2025

- Added legend slide
- Slide 16: Added antineoplastics
- Re-formatted drug charts throughout for visibility

June 2025

- Removed mention of folic acid 5 mg with carbamazepine and valproic acid from speaker notes as differs from SOGC recommendations

August 2025

- Formatting and grammar changes throughout
- Slide 2: Added NAPRA competencies
- Slide 3: Updated legend
- Removed Table of Contents slide
- Slide 11: Added note for SMX/TMP
- Slide 17: Updated hyperlinks
- Slide 18: Added takeaway slide
- Slides 19, 20: Added case study and questions

October/November 2025

- Slide 17: Changed 'RxTx' to 'Compendium of Pharmaceuticals and Specialties (CPS)'

March 2026

- Slide 2: Updated NAPRA competencies
- Formatting changes made throughout
- Slide 5: Changed kDa to Da
- Slide 16: Added radiopharmaceuticals next to iodine
- Slide 17: Added ® where applicable; minor edits made